



Smart energy for smart buildings

Product walkthrough — page by page

Live platform screens + real report outputs · for showing at metiundo, 1 July 2026

What you're flipping through

An estimation-first energy & compliance platform for building portfolios. It starts from data a building already has (bills, meters, EPCs) with no hardware, and every page unlocks more as you connect data and devices. These pages recreate the live app faithfully with real sample-portfolio numbers, followed by two genuine PDF report outputs.

In this deck

- Portfolio overview
- Embedded Power BI report
- Compliance (EPBD / CRREM / ESRS)
- Financing & subsidies
- Connections (hardware)
- Real reports: CRREM + GHG
- Buildings & building detail
- Reports & documents hub
- Decarbonisation & recommendations
- Alerts
- Copilot
- How the numbers & software are built

Portfolio

Portfolio Overview · 10 buildings · last 30 days

Your whole portfolio at a glance — energy intensity (EUI), cost and carbon per building, worst performers first. Click any building to drill into its reports, alerts and recommendations.

MIXED · 10 of 15 are demo

TOTAL ENERGY

2.14M kWh

▼ 22.3% vs prior 30d

AVG EUI

296.4 kWh/m²/yr

▼ 22.3% vs prior 30d

TOTAL COST

€443.21K

▼ 21.5% vs prior 30d

TOTAL CO₂

748.2K kg CO₂e

▼ 22.0% vs prior 30d

On-site Solar last 30 days

SOLAR GENERATED

124.51K kWh

▲ 21.4% vs prior 30d

RENEWABLE RATE

3.2 %

▲ 38.6% vs prior 30d

SOLAR EXPORTED

55.99K kWh

▲ 43.7% vs prior 30d

CO₂ AVOIDED

21.24K kg CO₂e

▲ 2.1% vs prior 30d

Buildings ranked — worst EUI first

B002 Istanbul Ticaret Merkezi	Retail	≈ 640		EPC E
B004 Wien Grand Hotel Delta	Hotel	≈ 480		EPC C
B005 Frankfurt Klinikum Epsilon	Healthcare	≈ 410		EPC C
B003 Hamburg Logistics Hub	Logistics	≈ 130		EPC A
B001 Berliner Bürogebäude Alpha	Office	≈ 48		EPC B

EUI (kWh/m²·yr) vs portfolio

Buildings

Buildings · 15 buildings in your portfolio

Sample buildings carry full demo data — useful for previewing the reports. Tap any building to open its report.

OFFICE

SAMPLE

B001

Berliner Bürogebäude Alpha

Berlin

5,200

m² · built 2005

RETAIL

SAMPLE

B002

Istanbul Ticaret Merkezi Beta

Istanbul

8,500

m² · built 1998

LOGISTICS

SAMPLE

B003

Hamburg Logistics Hub Gamma

Hamburg

12,000

m² · built 2018

HOTEL

SAMPLE

B004

Wien Grand Hotel Delta

Vienna

7,800

m² · built 2008

HEALTHCARE

SAMPLE

B005

Frankfurt Klinikum Epsilon

Frankfurt

15,000

m² · built 2000

EDUCATION

SAMPLE

B006

Amsterdam Universiteit Zeta

Amsterdam

9,500

m² · built 1985

OFFICE

SAMPLE

B007

Copenhagen Net-Plus HQ Eta

Copenhagen

6,500

m² · built 2023

OFFICE

SAMPLE

B008

Leipzig Plattenbau Theta

Leipzig

9,200

m² · built 1968

B001 — Berliner Bürogebäude Alpha

• **Berliner Bürogebäude Alpha** · DE · Berlin

Office · EPC B · Smart building · Sample read-only · 5,200 m² · built 2005

ENERGY INTENSITY (EUI)

48 kWh/m²·yr

annualised run-rate

ENERGY USED · 30D

20,350 kWh

total, last 30 days

ENERGY COST · 30D

€3,372

estimated

CARBON · 30D

5,669 kg CO₂e

Scope 2 grid

OPEN ALERTS

0

high + critical

SAVINGS OPPORTUNITIES

7

open recommendations

MODULES

Meters

IoT

Battery

Open for this building

- Energy report — full Power BI
- Alerts — 0 open
- Recommendations — 7 to review
- Solar · Export PDF (building report)

📌 **AI Energy Advisor**

Reading this building's live data

7 savings opportunities to review

Biggest: Improve Building Insulation — Wall & Roof (~€3,999/yr, ~13 yr payback). EUI below typical band (100–200 kWh/m²·yr, office) — GOOD.

Reports & documents → everything generated for this building on one page (next)

Every building is the drill-down hub: KPIs, the embedded Power BI report, alerts, ranked recommendations, solar, and one-click PDF exports of the compliance & ESG documents.

B001 · Portfolio Overview

• Embedded Power BI report · DE · Office · 5,200 m²

REPORTS

Portfolio Overview

Building Detail

Anomalies & Alerts

Forecast & Recommend.

Occupancy Analysis

Sustainability Comp.

HVAC Analysis

IoT Monitoring

Battery Strategy

Solar Performance

MONITORING

Alerts

DOCUMENTS

Export PDF

Power BI

Interactive embedded report

(live report renders here — portfolio KPIs, anomalies, forecast, HVAC, occupancy)

How to read this

EUI benchmark by building

Energy per m²·yr vs the portfolio median.

Consumption trend vs PY

Rising while floor area is flat = creeping waste.

Cost & CO₂ totals

CO₂ drives CRREM/EPC risk even when the bill looks fine.

Building summary table

Per-building EUI, cost & class — triage first.

Locked: IoT & Solar pages unlock with hardware modules

B001 · Reports & documents

• Reports & documents · pick a report to open it

UNLOCK MORE — completeness score grows as you add data

- +30 Add 12 months of energy use → unlocks Energy KPIs & EUI, GHG Inventory, ESRS-E1, VSME, GRESB, CR
- +15 Add Envelope U-values
- +12 Add Heating system & fuel → unlocks CO₂ Cost Split

DASHBOARDS

Energy dashboards

Interactive Power BI — portfolio, anomalies, forecast & HVAC. 2 pages need dashboard model

DASHBOARDS

Open →

BUILDING DOCUMENTS German-mandatory screening

CO₂ Cost Split

Landlord/tenant CO₂ allocation under CO₂KostAufg (est., area pro-rata).

BUILDING

Open →

GEG conformity

Heating (\$71) & envelope U-value screening under the Building Energy Act.

BUILDING

Open →

EPC pre-assessment

Indicative Energieausweis class — screening, not a registered certificate.

BUILDING

Open →

PORTFOLIO ESG — organisation-wide reports

ESRS E-1 (Climate)

Climate disclosure E1-1...E1-9, ESRS-aligned (Scope 1/2/3 auto-filled).

PORTFOLIO

Open →

GHG Inventory

Scope 1/2/3 greenhouse-gas inventory in GHG Protocol framing.

PORTFOLIO

Open →

EnEfG plan

Energy-efficiency measures plan (Maßnahmenplan).

PORTFOLIO

Open →

VSME report

Voluntary SME sustainability standard summary.

PORTFOLIO

Open →

CRREM stranding

1.5°C transition-risk stranding year per asset.

PORTFOLIO

Open →

GRESB readiness

Real-estate ESG benchmark readiness snapshot.

PORTFOLIO

Open →

Compliance

• **Compliance & Sustainability**

· EPBD/MEPS · CRREM · flexibility · EU Taxonomy · ESRS-E1

How your portfolio stands against the EU rules that increasingly drive building value — MEPS renovation risk, CRREM stranding, EU Taxonomy and ESRS-E1. Indicative screening, not a legal verdict.

Compliance snapshot

10 buildings screened — indicative, not a legal verdict.

● **RENOVATION PRIORITY**

1

G→2030:1 · F→2033:0

● **STRANDED NOW**

5

5 strand by 2030 · 10/10 assessed

● **TAXONOMY A-ROUTE**

3/10

26% area

● **FLEXIBILITY-READY**

4

6 partial · 0 limited

● **ESRS-E1**

Ready

reporting 2025

MEPS

CRREM

Flexibility

EU Taxonomy

Screening

ESRS-E1

Renovation & MEPS risk — worst-class buildings first

B002 Istanbul Ticaret Merkezi

EPC E

Must renovate before 2030

B004 Wien Grand Hotel Delta

EPC C

At risk by 2033

B005 Frankfurt Klinikum Epsilon

EPC C

At risk by 2033

B001 Berliner Bürogebäude Alpha

EPC B

On track

B003 Hamburg Logistics Hub

EPC A

Compliant

Decarbonisation

• Decarbonisation planning

· All buildings · measures ranked cheapest-first

Plan where to invest for the most CO₂ cut per euro. The curve ranks measures cheapest-first — green ones (below €0) pay for themselves.

ABATABLE CO₂**1,524** tCO₂/yr

43 measures

NO-REGRET CO₂**1,351** tCO₂/yr

measures that pay for themselves

NO-REGRET CAPEX

€3.29M

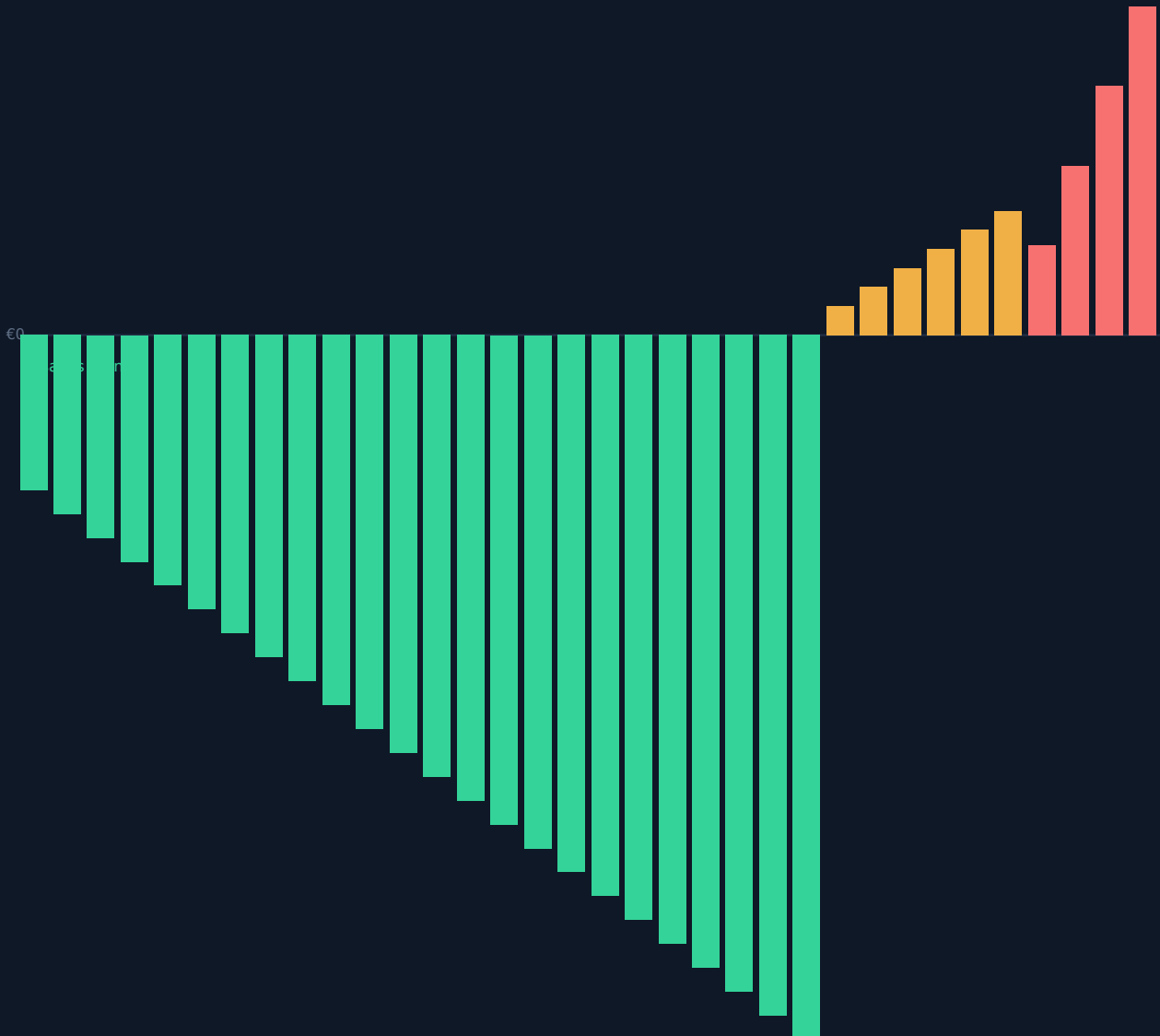
self-funding investment

WEIGHTED AVG COST

€-326 /tCO₂CO₂-weighted across portfolio

Marginal abatement cost curve

Each bar is one measure. Width = CO₂ cut per year; height = cost per tonne. Below the €0 line (green) the measure saves money over its life; above it (amber/red) it costs money. Cheapest-to-abate on the left.

€/tCO₂

cheapest to abate → most expensive

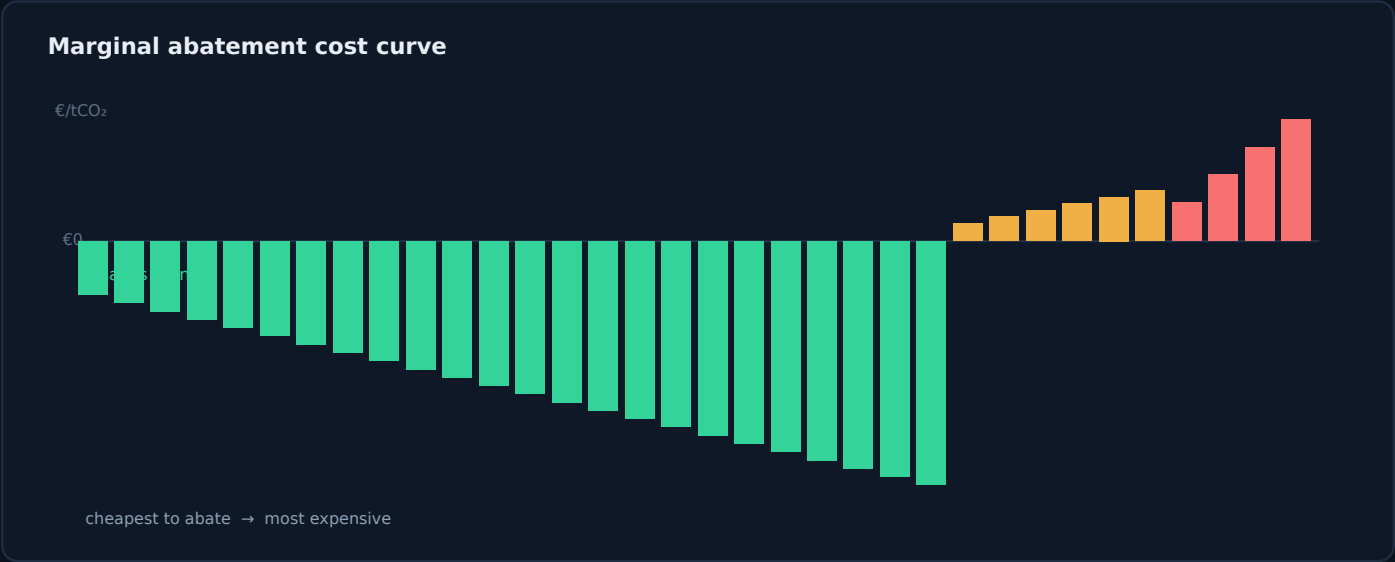
Actions

Recommendations · 57 recommendations across the portfolio

Recommended improvements with savings, cost and payback. Set each one's status to track progress. Open Decarbonisation for the CO₂ investment view, or filter by building.

33 no-regret · 1,365 tCO₂/yr

1,524 tCO₂/yr abatable



Top ranked measures

MEASURE	BUILDING	PROGRAMME	SAVING/YR	PAYBACK
Improve building insulation — wall & roof	B001	BAFA	€3,999/yr	~13 yr
Install heat pump (air-source)	B002	BEG/KfW	€12,400/yr	~9 yr
LED retrofit + controls	B004	—	€2,100/yr	~3 yr
CHP cogeneration unit	B005	KWKG	€48,000/yr	~7 yr
Rooftop PV + battery	B003	EEG/BAFA	€9,800/yr	~11 yr

Alerts

Alerts · 6 open issues across the portfolio

Automated fault detection across your buildings. Recurring anomalies are grouped into one ongoing issue — not one alert per day — so the list stays actionable.

UNHANDLED CRITICAL

0

needs attention now

UNHANDLED HIGH

6

investigate soon

OPEN & UNHANDLED

6

active triage queue

ACKNOWLEDGED

2

owned by a human

BUILDING	SEVERITY	ANOMALY	READING
Frankfurt Datacenter Iota B009	High	HIGH_CARBON_INTENSITY Peak carbon 1.9150 kg CO ₂ /m ² /day > 1.3× CRREM 2024	1.9 v +169% Open
Amsterdam Universiteit Zeta B006	High	HIGH_CARBON_INTENSITY Peak carbon exceeds education reference	0.2 v +52% Open
Frankfurt Klinikum Epsilon B005	High	HIGH_CARBON_INTENSITY Peak carbon exceeds healthcare reference	0.7 v +134% Open

Connections

• **Devices & Connections** · wire up live data sources

Devices start pending. An on-site edge agent or poller activates them once it reports in — the backend never reaches devices behind your firewall directly.

Schneider PowerLogic PM5560

BACnet/IP

PENDING

port 47808

Waiting for the on-site agent to report its first reading

AI:1	→	HVAC supply temp	°C	on
AI:2	→	HVAC return temp	°C	on
AI:3	→	Occupancy	ppm	on
AI:4	→	Humidity	%	on
+	→	Building power (total)	kW	on

Modbus meter #343

Modbus TCP

PENDING

unit 1

Waiting for the on-site agent to report its first reading

40001	→	Active power	kW	on
40003	→	Energy total	kWh	on
40007	→	Voltage	V	on

This is the hardware bridge

Supports BACnet/IP, Modbus TCP, MQTT and REST via an on-site edge agent. Connecting devices is what lights up the live-monitoring, HVAC efficiency and battery pages — the more you connect, the more you unlock.

Copilot

Copilot

· AI assistant for your portfolio

Start the conversation — or try one of these — or try one of these:

Which buildings have the worst EUI?

What's our total abatable CO₂ per year?

Show the no-regret measures that pay for themselves

B001 anomalies in the last 30 days

Ask about a building (e.g. 'B001 anomalies last 30 days')...

Send

Deterministic insights are grounded in the building's own data; the conversational copilot layers a language model on top for plain-English Q&A.

EnergyLens

Real report outputs

The next pages are genuine PDF documents the platform generates — not recreations. CRREM stranding assessment and a full GHG Protocol inventory, exported straight from a building.

EnergyLens

Smart energy for smart buildings

CRREM — Stranding assessment

Generated 29 June 2026 at 14:06 UTC

Berliner Bürogebäude Alpha - single building

CRREM-aligned stranding

Indicative 1.5°C pathways — not the licensed CRREM dataset

Transition-risk screening using the CRREM **method**: a building's annual operational carbon intensity is compared against a declining 1.5°C decarbonisation pathway, and the first year the (held-flat) intensity exceeds the pathway is its **stranding year**. The pathways here are indicative, 1.5°C-aligned anchors by asset type — **not** the official licensed CRREM curves (which require a CRREM License Partner agreement) and not country-grid specific. Decision-support, not an assured disclosure.

PORTFOLIO STRANDING SUMMARY

ASSESSED

1/1

buildings with data

STRANDED NOW

0

above today's pathway

STRANDING BY 2030

0

cumulative

STRANDING BY 2035

0

cumulative

AVG STRANDING YEAR

2045

of stranded assets

PER-ASSET ASSESSMENT

BUILDING	INTENSITY	PATHWAY 2026	STRANDING YEAR	STATUS
Berliner Bürogebäude Alpha OFFICE	13.3	43.4	2045	Stranding 2045

Intensity and pathway are in kgCO₂/m²-yr. Intensity is annualised from the latest ~30 days of operational emissions and held flat — a conservative screening view that does not credit planned decarbonisation. Improving efficiency or electrifying heat (see the recommendations) pushes the stranding year out.

METHOD & ASSUMPTIONS

Pathways are linearly interpolated between indicative 2025 and 2050 anchors per asset type (1.5°C-aligned). A building is **stranded now** if its intensity already exceeds the 2025 pathway, **on track** if it stays at or below the 2050 endpoint, otherwise **stranding** in the first year the pathway falls below it. Swapping in the licensed CRREM curves (per type and country grid) is a single-module change once a License Partner agreement is in place.

Prepared using the CRREM methodology with indicative 1.5°C pathways. This is reporting support / transition-risk screening — it is not the official CRREM tool output, not audited or assured, and not investment advice. Confirm against the licensed CRREM dataset before external use.

energylens.eu · Confidential

CRREM-aligned (indicative pathways) · reporting support (not audited)

Generated 29 June 2026 at 14:06 UTC

EnergyLens · visual walkthrough (recreated from the live app, real data) · metiundo, 1 Jul 202615 / 18

EnergyLens

Smart energy for smart buildings

GHG Inventory — GHG Protocol

Generated 29 June 2026 at 14:05 UTC

Berliner Bürogebäude Alpha · single building · 2025

GHG Protocol Corporate Standard

Scope 1+2 disclosure-grade · Scope 3 estimated

Reporting year 2025

INVENTORY SCOPE & BOUNDARY

Reporting entity: portfolio of 1 building (5,200 m²). **Organizational boundary:** operational control (buildings the organization operates). **Reporting period:** 2025. **Scope 2 reporting:** dual — location-based and market-based, per the GHG Protocol Scope 2 Guidance. **Base year:** 2025 (first complete reporting year; recalculate on a structural change ≥ 5%). **Gases & GWP:** the Kyoto-basket gases reported as CO₂e using IPCC AR6 GWP-100; building emissions are predominantly CO₂. **Consolidation:** 100% of operated assets (operational-control approach).

OPERATIONAL CARBON — SCOPE 1 + 2 (TCO₂E)

OPERATIONAL TOTAL

121.6

Scope 1 + 2 — building energy

OPERATIONAL INTENSITY

23.4

kg CO₂e/m²·yr — EPBD / CRREM basis

Operational carbon is the building's own energy footprint (on-site combustion + purchased electricity) — the basis for EPBD and CRREM. The full inventory below adds estimated value-chain (Scope 3) emissions such as employee commuting and business travel.

EMISSIONS BY SCOPE (TCO₂E)

SCOPE 1 — DIRECT

3.5

on-site fuel combustion

SCOPE 2 — LOCATION

118.1

grid-average electricity

SCOPE 2 — MARKET

235.7

contractual / supplier

SCOPE 3 — VALUE CHAIN

698.7

estimated value chain — incl. commuting & travel

TOTAL (LOCATION-BASED)

820.3

Scope 1 + 2(loc) + 3

TOTAL (MARKET-BASED)

937.9

Scope 1 + 2(mkt) + 3

GHG INTENSITY

0.158

tCO₂e/m² — incl. value chain

RENEWABLE ELECTRICITY

16.8 %

on-site solar self-consumed

EnergyLens · visual walkthrough (recreated from the live app, real data) · metiundo, 1 Jul 2026

16 / 18

WHAT EACH SCOPE COVERS HERE

SCOPE	INCLUDED	GRADE
Scope 1	Direct on-site combustion — natural gas + heating oil/diesel	Disclosure-grade
Scope 2	Purchased electricity — location-based (grid factor) and market-based (supplier / Guarantees of Origin)	Disclosure-grade
Scope 3	Value chain — screening estimate spanning purchased goods/embodied (Cat 1), fuel & energy (Cat 3), waste (Cat 5), business travel (Cat 6), employee commuting (Cat 7) and downstream leased assets (Cat 13).	Estimated — not disclosure-grade

FACILITY BREAKDOWN (TCO₂E)

BUILDING	SCOPE 1	SCOPE 2 (LOC)	SCOPE 2 (MKT)	SCOPE 3	TOTAL (LOC)	INTENSITY
Berliner Bürogebäude Alpha OFFICE	3.5	118.1	235.7	698.7	820.3	0.158

DATA QUALITY & METHODOLOGY

BUILDINGS REPORTED

1 / 1

with GHG data

COMPLETE

0

metered

ESTIMATED

0

modelled

MISSING GAS

1

needs fuel data

Prepared with reference to the GHG Protocol Corporate Accounting and Reporting Standard and the Scope 2 Guidance (location- and market-based). Emission factors: national grid factors (Scope 2 location), supplier/GoO factors where available (Scope 2 market), BEHG/standard fuel factors (Scope 1), each at the most recent published vintage for the reporting year (state the exact factor source year in the final report). Scope 3 is a screening estimate spanning categories 1, 3, 5, 6, 7 and 13 (incl. business travel and employee commuting) and is not disclosure-grade; biogenic CO₂ and removals are excluded. This inventory is reporting support — it has not been independently verified or assured.

energylens.eu · Confidential

Scope 1+2 disclosure-grade · Scope 3 estimated

Generated 29 June 2026 at 14:05 UTC

"How did you calculate this — based on what?"**Say (simple):**

"Every number comes from standard, published methods, not guesses. Energy intensity is measured energy divided by floor area. Carbon uses official grid emission factors. CRREM stranding compares each building to the EU 1.5°C decarbonisation pathway. Retrofit savings use engineering formulas — heat loss from U-values, area and degree-days, divided by system efficiency. Subsidies use current German rates like BAFA and KfW. And I always show a range with the assumptions, never one false-precise number."

One line:

"Standard energy-engineering and EU methodology — EUI, grid factors, CRREM pathways, degree-day heat models — always shown as ranges with clear assumptions."

"How did you build it — the data and software?"**Say (simple):**

"It runs on Microsoft Fabric. Raw data lands in a Bronze layer, gets cleaned in Silver, and the business logic sits in a Gold layer — the medallion pattern. On top of that a web app serves the dashboards: Next.js front end, Python/FastAPI back end, a Postgres database, and Power BI embedded for the detailed reports. Today it runs on synthetic sample buildings that mirror real EU building types; the next step is real pilot data. I built the whole thing solo."

One line:

"Microsoft Fabric with a Bronze/Silver/Gold data model, a Next.js + FastAPI + Postgres web app, and embedded Power BI. Sample data now, real pilots next. Built solo."

Honesty anchor:

"Built and live, lining up first pilots. Solo for now, incorporating as pilots land." — never invent customers or revenue.